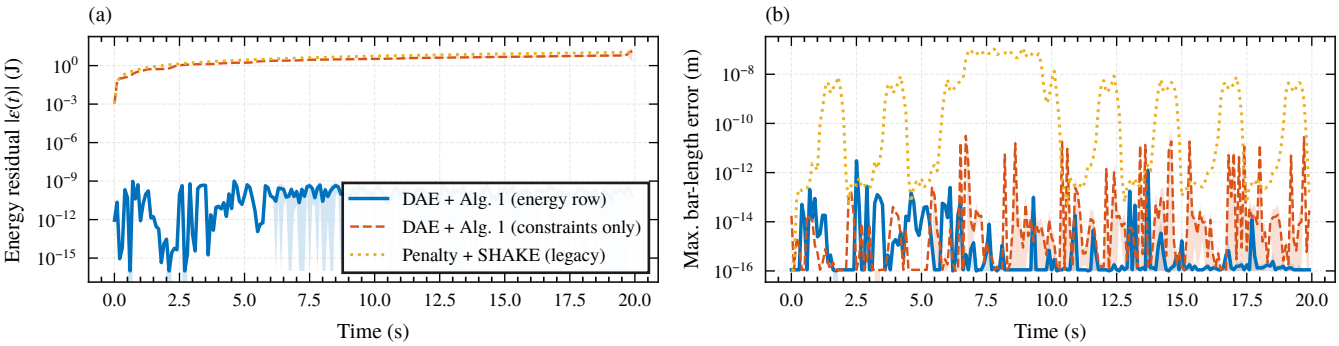
(a)
Energy residual |e(t)| (J)
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)
DAE + Alg. 1 (energy row)
DAE + Alg. 1 (constraints only)
Penalty + SHAKE (legacy)
(b)
Max. bar-length error (m)
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)